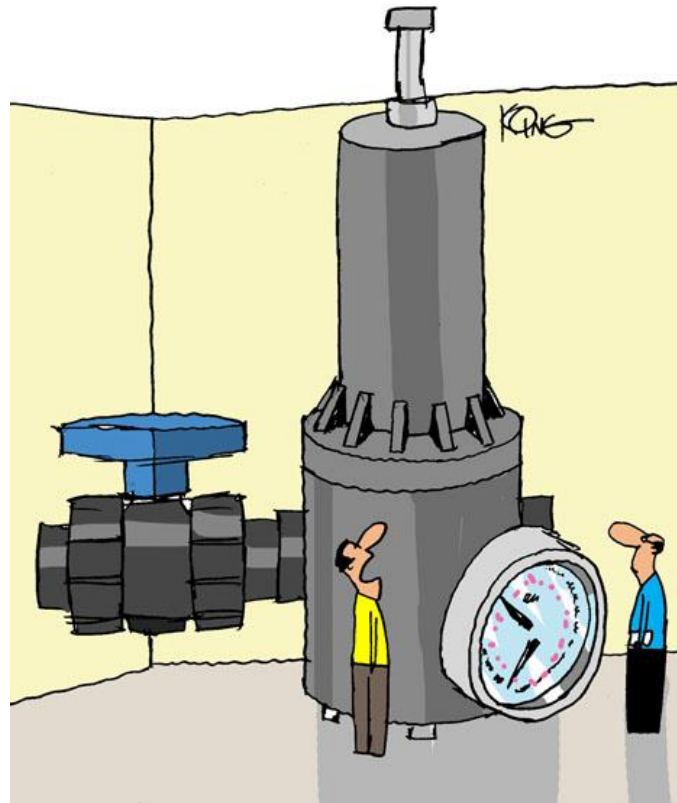


Identification - Introduction

Radoslav Paulen



1 The accelerating pace of change ...



Moore's Law

2 ... and exponential growth in computing power ...

Computer technology, shown here climbing dramatically by powers of 10, is now progressing more each hour than it did in its entire first 90 years

COMPUTER RANKINGS

By calculations per second per \$1,000



Analytical engine

Never fully built, Charles Babbage's invention was designed to solve computational and logical problems



Colossus

The electronic computer, with 1,500 vacuum tubes, helped the British crack German codes during WW II



UNIVAC I

The first commercially marketed computer, used to tabulate the U.S. Census, occupied 943 cu. ft.



Apple II

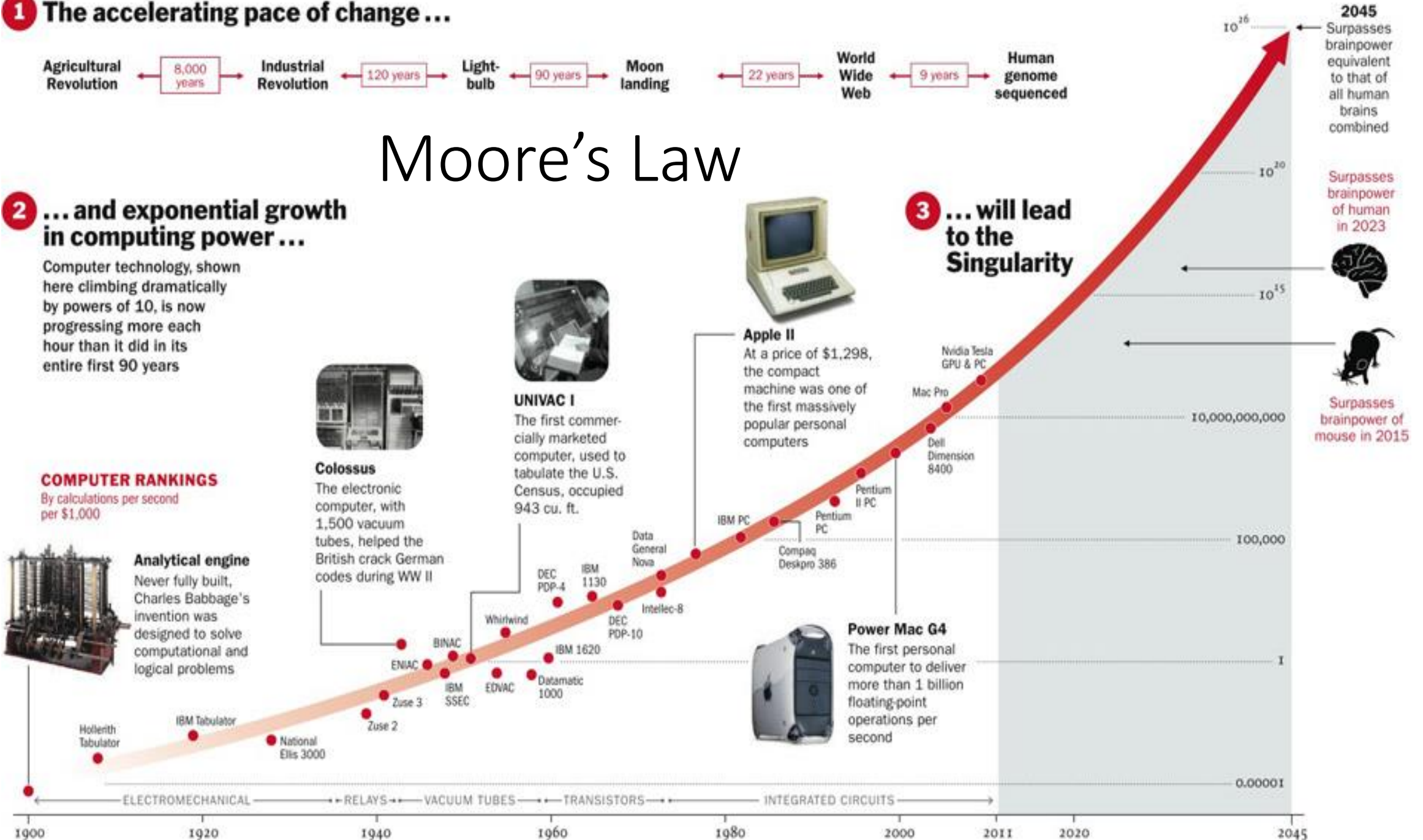
At a price of \$1,298, the compact machine was one of the first massively popular personal computers



Power Mac G4

The first personal computer to deliver more than 1 billion floating-point operations per second

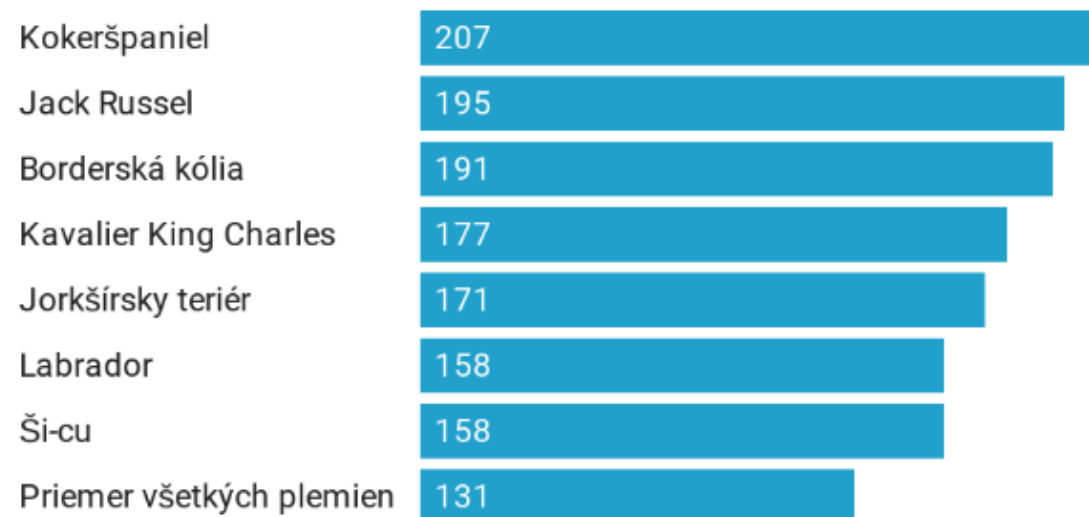
3 ... will lead to the Singularity



The world is extremely rich in data

Ceny čistokrvných psů v Británii počas pandémie výrazne stúpli

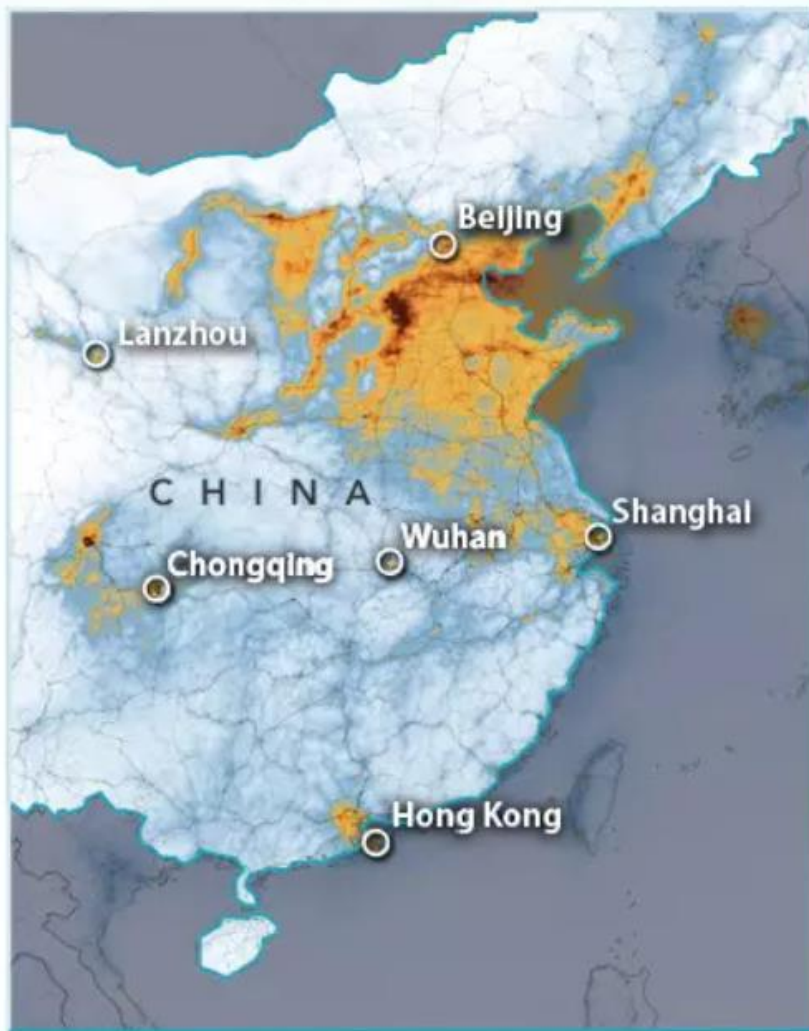
(medziročný rast cien v %)



Zdroj: Pets4Homes.co.uk •
Vytvorené pomocou Datawrapper

NO_2 levels also plummeted in the rest of China as large parts of the country were shut down in an effort to contain the spread of COVID-19.

Jan 1–20, 2020



Feb 10–25, 2020



Source: NASA Earth Observatory

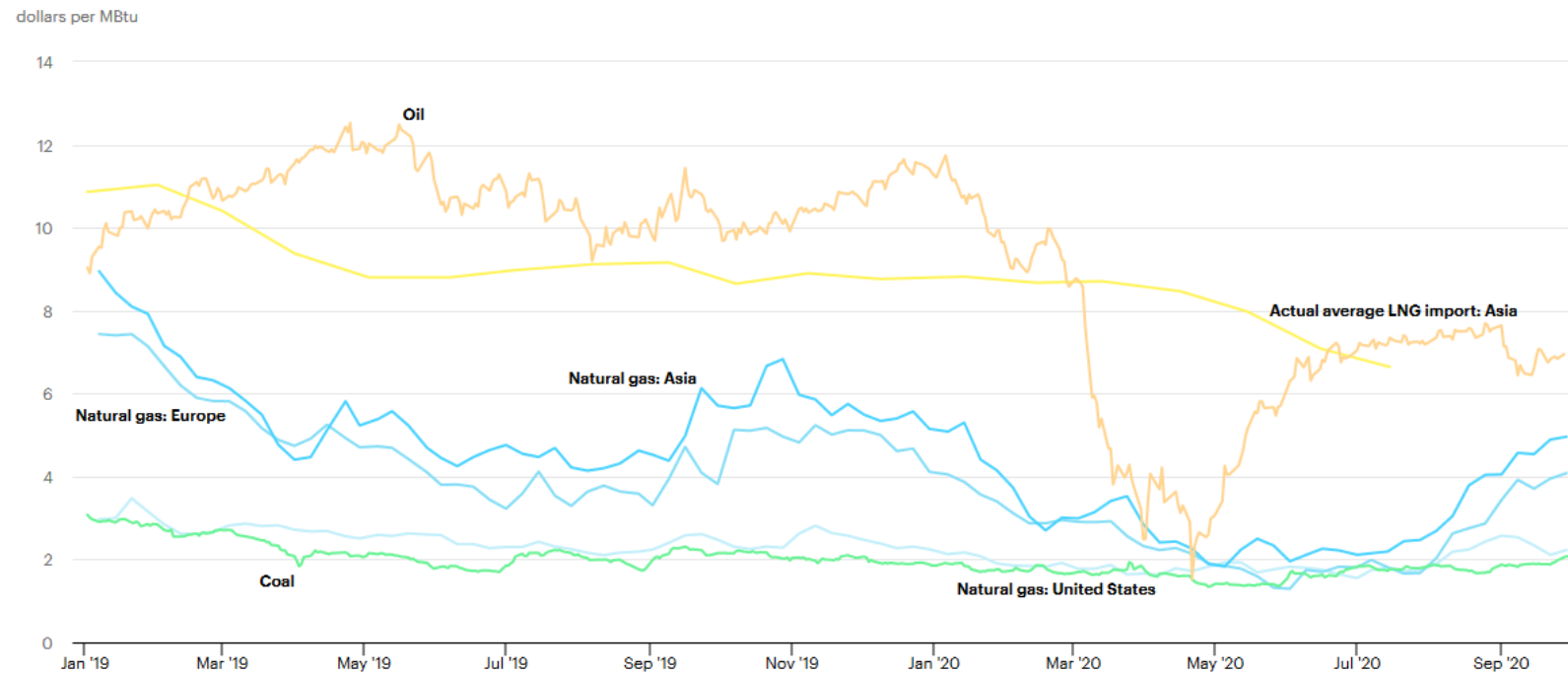


Selected fossil fuel prices in 2019-2020

ast updated 12 Oct 2020

Download chart ↓

Cite Share



IEA. All Rights Reserved

Oil Coal Natural gas: Asia Natural gas: Europe Natural gas: United States Actual average LNG import: Asia

The Kyoto Aquarium has a flowchart illustrating the complicated romantic relationships and breakups between their penguins.



Facebook na niekoľko týždňov zablokoval stránky francúzskeho mesta Bitche, pretože si jeho názov pomyšľal s anglickou nadávkou. Hoci sa 5-tisícové mesto snažilo spoločnosť kontaktovať od 19. marca, chybu sa podarilo napraviť až teraz.

Algoritmus sociálnej siete usúdil, že meno mesta porušuje pravidlá o užívaní vulgarizmov, pretože ho zamenil so slovom „bitch“, čo je v angličtine často používaná urážka.

Teraz sa šéf francúzskeho zastúpenia spoločnosti Facebook starostovi mesta za kurióznú zámenu ospravedlnil.

„Facebook nás 19. marca informoval, že naše stránky Ville de Bitche už nie sú on-line prístupné pre porušenie podmienok používania služby,“ uviedol starosta mesta Benoit Kieffer. „Zdá sa, že meno nášho mesta si niekto zle vyložil,“ dodal.

Podľa starostu tento prípad dokazuje, že metódy, ktoré Facebook uplatňuje pre dohľad nad obsahom, celkom nefungujú.

Než sa chybu podarilo napraviť, mesto Bitche na sociálnej sieti vystupovalo pod neškodným označením „Mairie 57230“, teda označením podľa poštového smerovacieho čísla. (**guardian**, **bbc**, **čtk**)



Google začne vo svojej aplikácii Google Maps navrhovať vodičom najekologickejšiu trasu. Aplikácia bude túto trasu odhadovať na základe rôznych faktorov vrátane hustoty premávky či výškového profilu trasy. (reuters)



Posted by u/[deleted] 5 years ago



15



Spotify Hackers Threaten to Reveal Full List of Users Who Listen to Nickelback

i.imgur.com/CZKLSd...



Stories Politics Video Memes Thoughts

United States

World News

Cincinnati

LATEST

Spotify Hackers Threaten to Reveal Full List of Users Who Listen to Nickelback

By Jesse McLaren August 21, 2015



New York, NY- The stock market remains in chaos as a hacker group makes yet another threat to reveal sensitive information from music streaming service Spotify's databases. This comes one week after the



PHOTO: Nickelback posing in truckstop bathroom.

2.1. 15:07

Zdieľať ➔

Cena bitcoinu prekonala hranicu 30-tisíc dolárov. Za rok sa cena kryptomeny zvýšila o vyše 300 percent. Bitcoin v poslednom čase ťaží najmä z dopytu zo strany väčších investorov, výrazne mu pomáha aj politika centrálnych bánk a tlač ďalších peňazí na boj proti pandémiei. (čtk, dpa)

11.1. 20:14

Zdieľať ➔

Po prudkom raste klesla cena bitcoinu za tri dni o viac než pätinu.

Ešte v piatok vzrástla cena kryptomeny na nový rekord takmer 42 000 dolárov. V pondelok to bolo už len približne 32 000 dolárov.

Čítať ďalej 102 slov

5.1. 17:37

Zdieľať ➔

Cena bitcoinu by sa mohla v dlhodobejšom horizonte vyšplhať až na 146 000 dolárov, uviedli analytici americkej investičnej banky JP Morgan. V súčasnosti stojí bitcoin asi 30 000 dolárov, jeho cena sa za dva týždne zvýšila o 50 %.

Čítať ďalej 123 slov

INVESTOVANIE, ÚVERY, POISTENIE >

22.1. 09:50

Zdieľať ➔

Hodnota bitcoinu sa dnes prepadla pod 30-tisíc dolárov a mieri k najväčšiemu týždennému poklesu od septembra. Na rekordnom maxime bola kryptomena pred dvomi týždňami. Investori sa obávajú novej regulácie.

Čítať ďalej 233 slov

7.1. 20:05

Zdieľať ➔

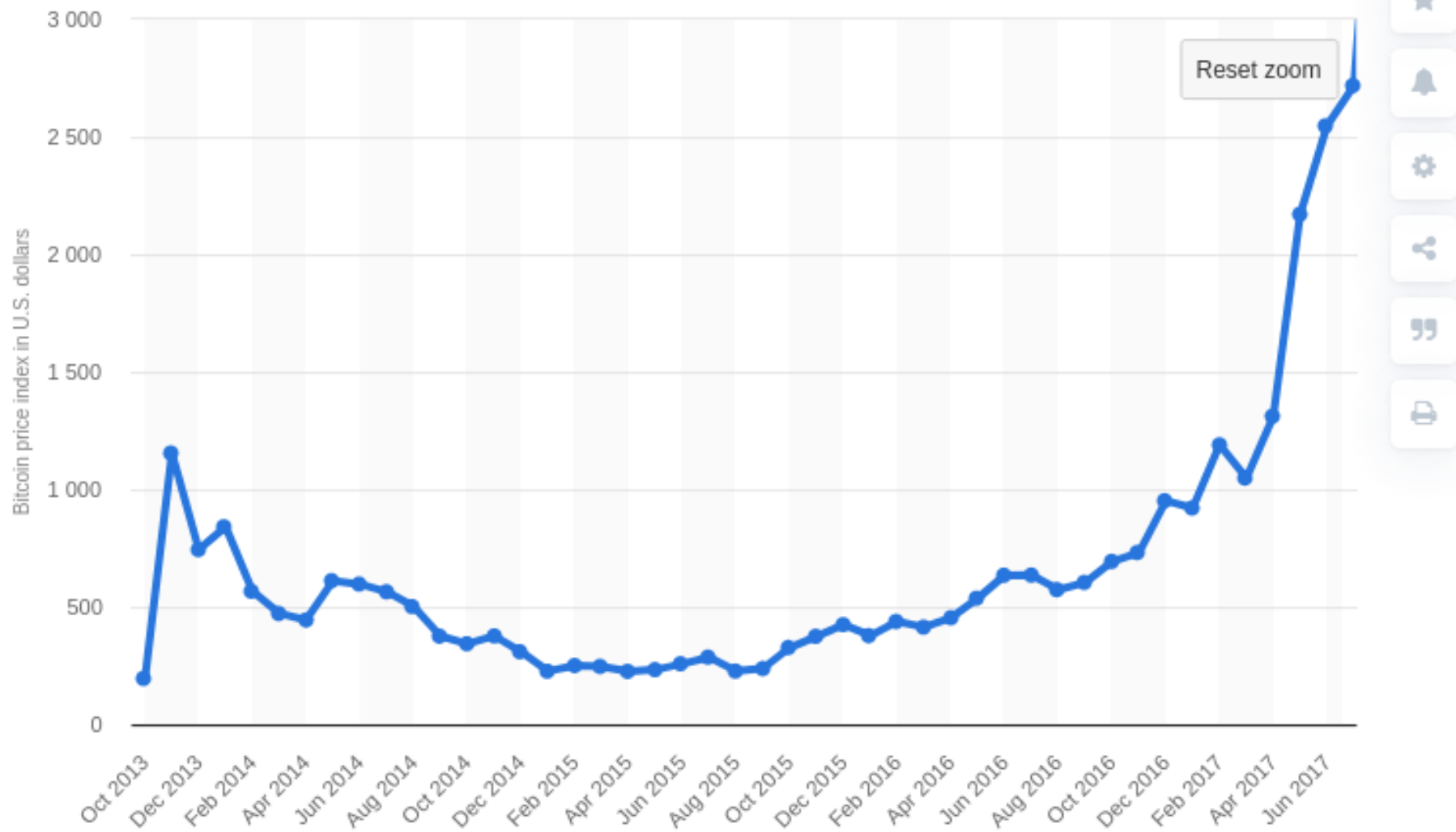
Cena bitcoinu stále stúpa, dnes prekonala hranicu 40-tisíc dolárov. Tejto kryptomene v súčasnosti pomáha dopyt zo strany väčších investorov, politika centrálnych bánk a tlačenie ďalších peňazí na boj s koronakrízou, čo vyvoláva obavy rýchleho rastu inflácie. (čtk, coindesk)

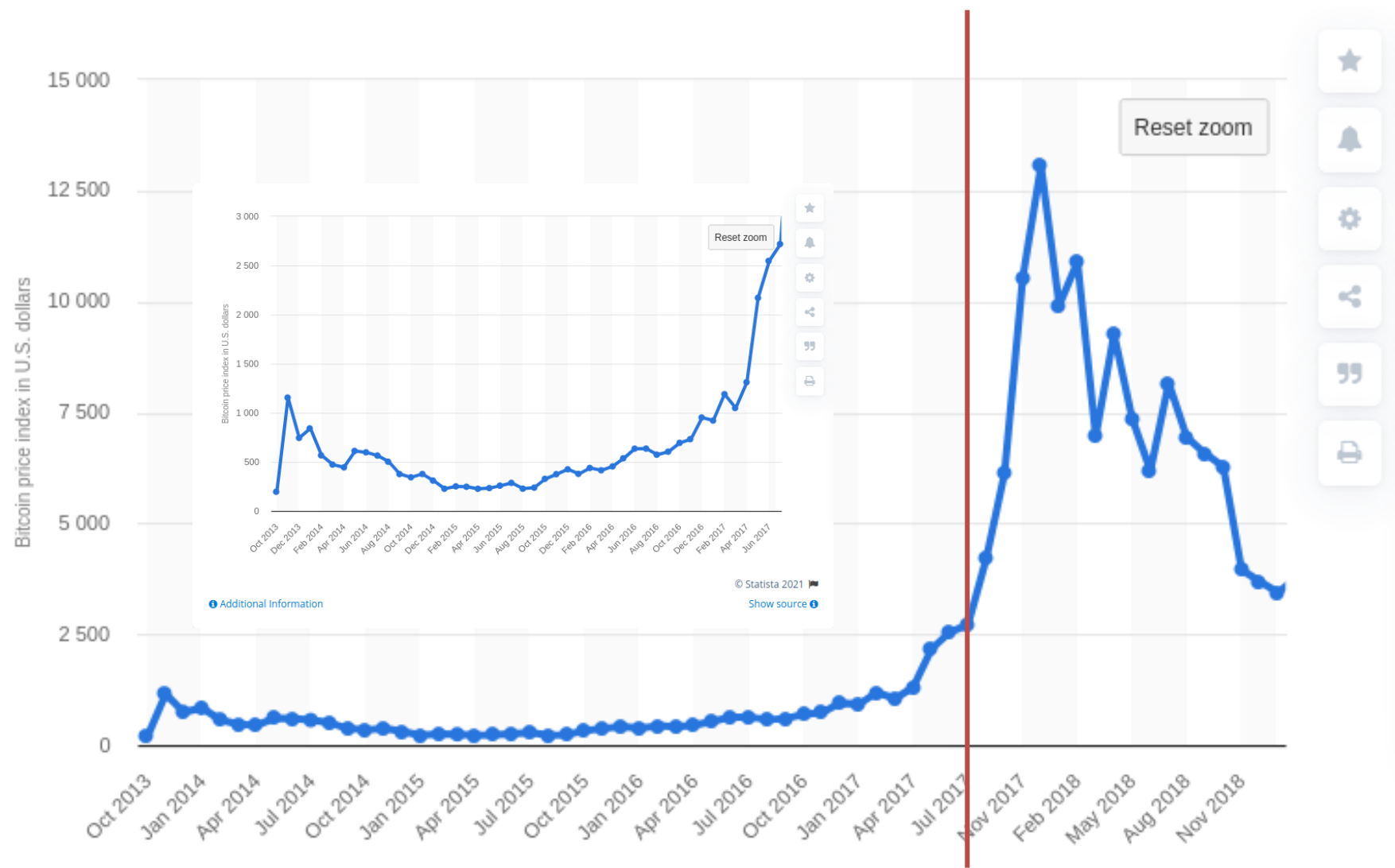
29.1. 15:18

Zdieľať ➔

Hodnota bitcoinu na burze Bitstamp sa dnes zvýšila o 20 % na približne 38-tisíc dolárov. Podľa analytikov k tomu prispel Elon Musk, keď na svoj twitterový účet umiestnil hashtag #bitcoin so zašifrovaným komentárom.

Čítať ďalej 119 slov

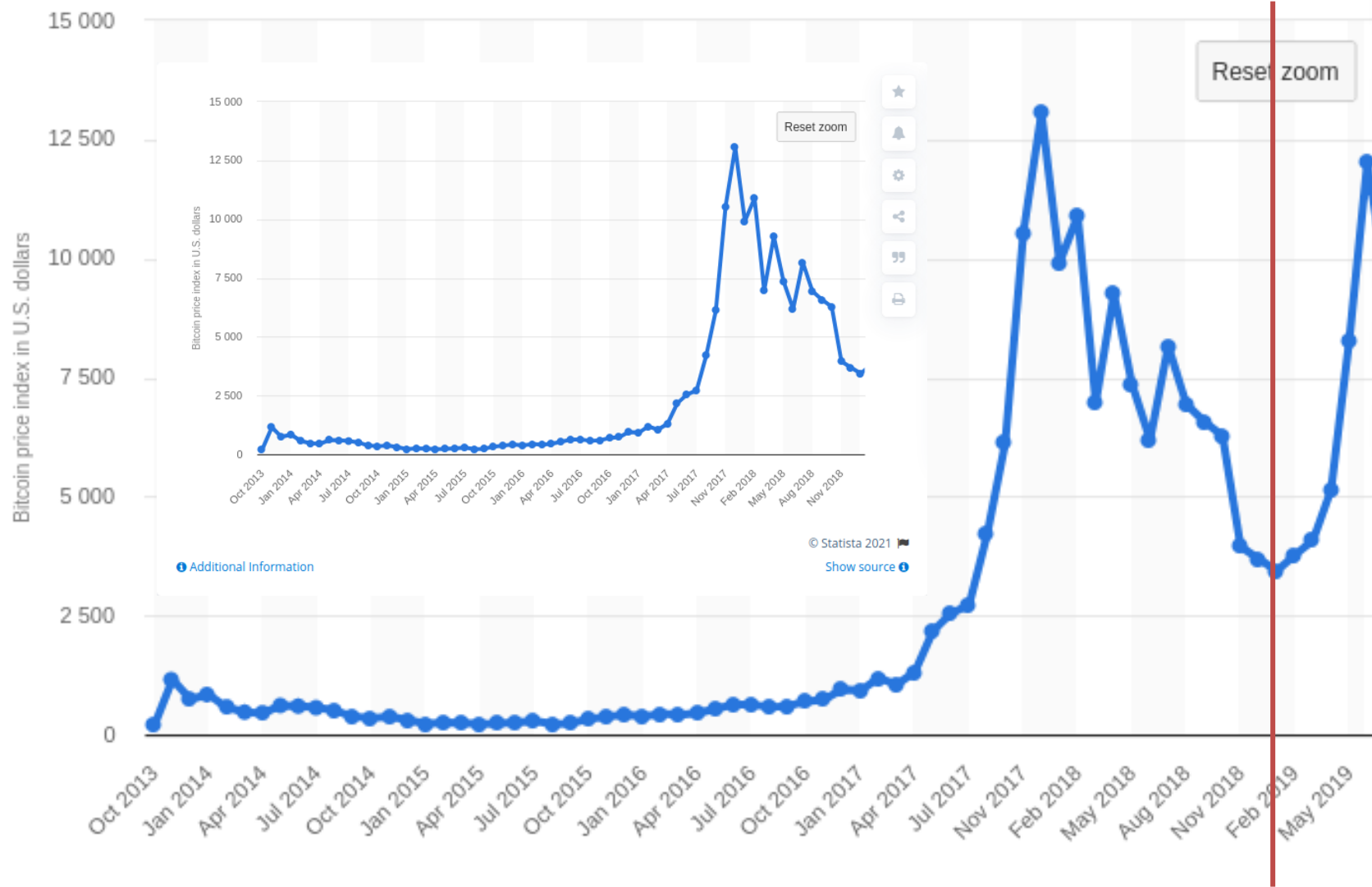




© Statista 2021

[Additional Information](#)

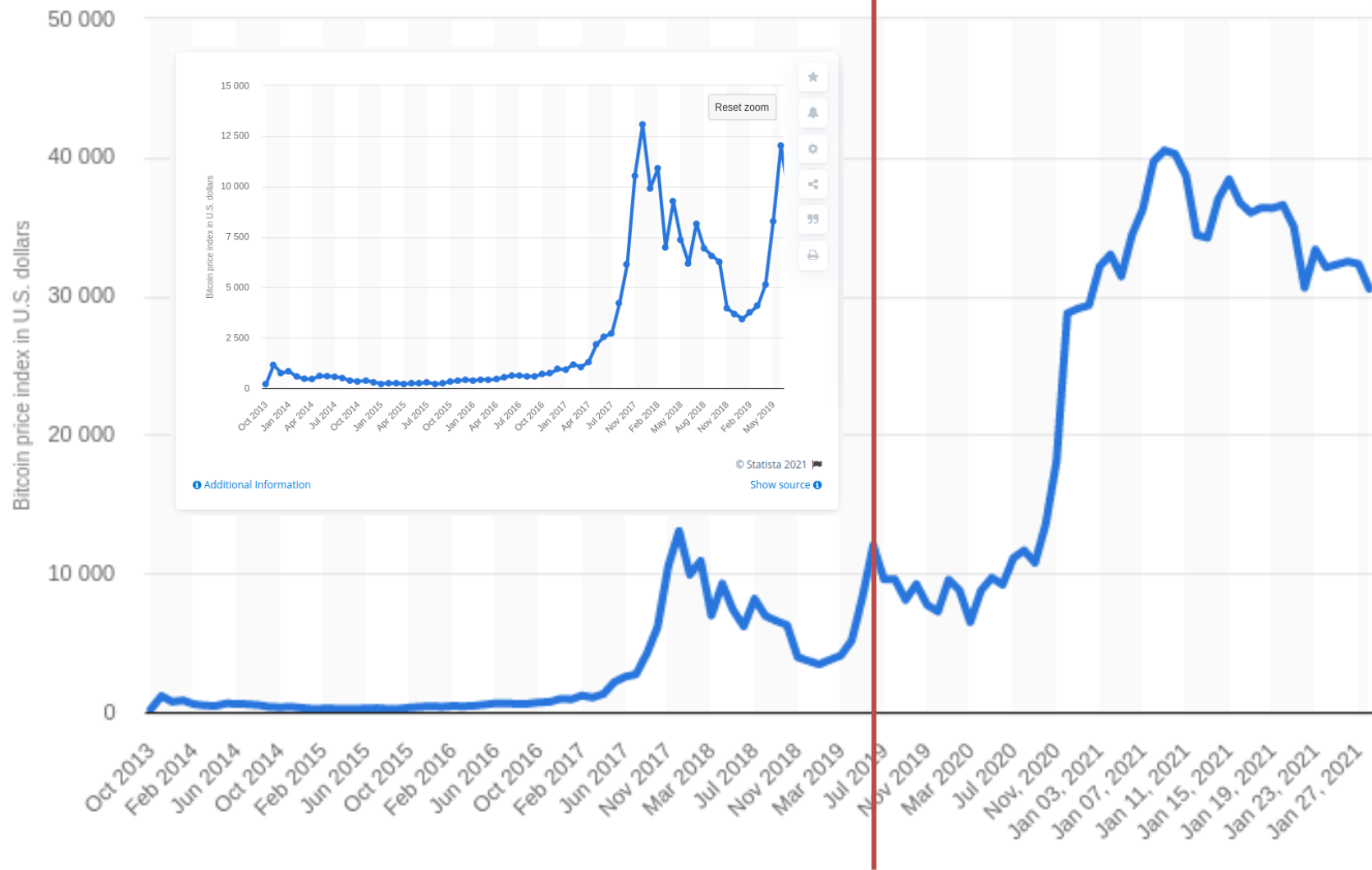
[Show source](#)



[Additional Information](#)

© Statista 2021

[Show source](#)

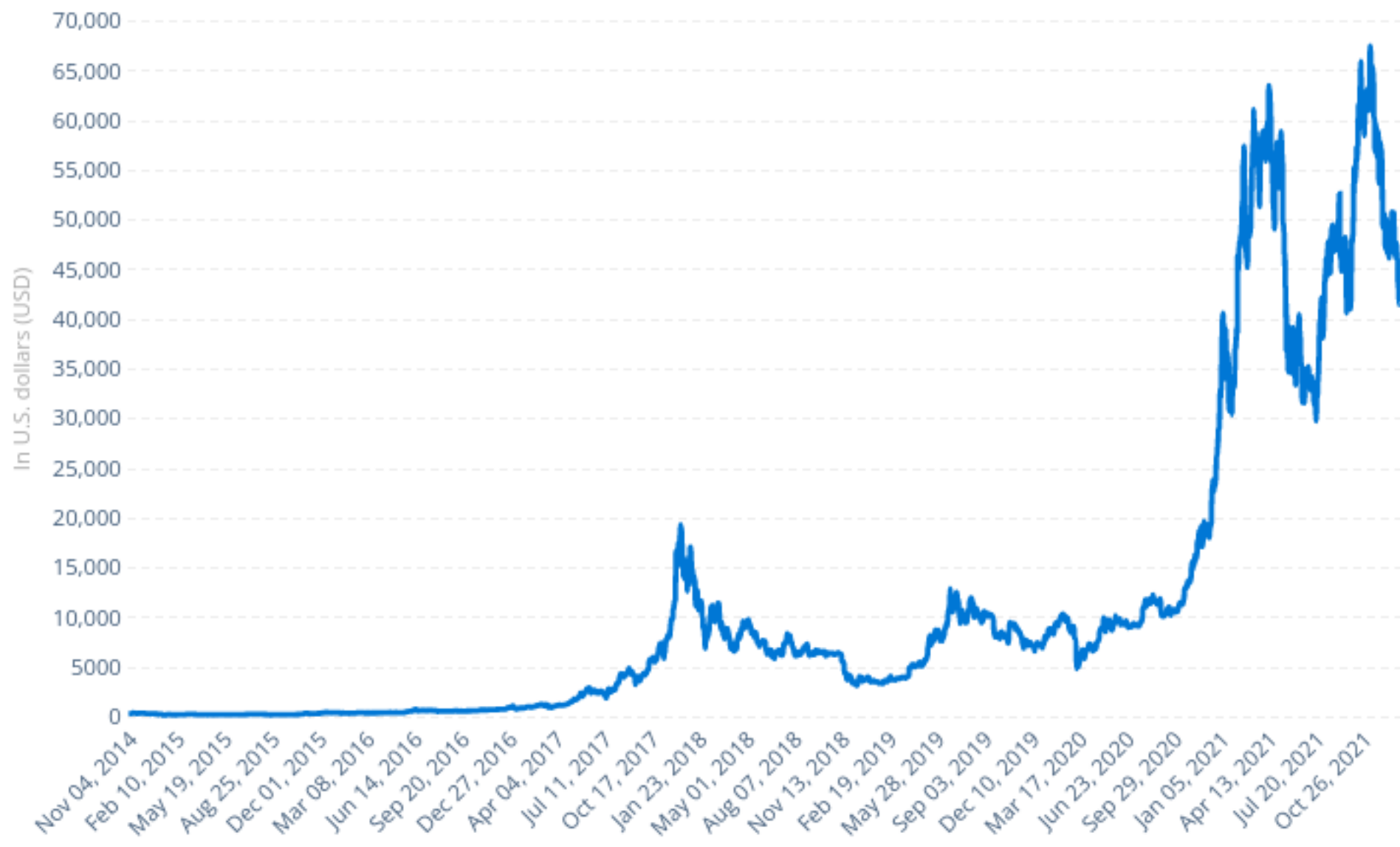


- ★
- 🔔
- ⚙️
- 🔗
- 🗣️
- 🖨️

Bitcoin (BTC) price history

Linear

Log



Bitcoin (BTC) price history

Linear

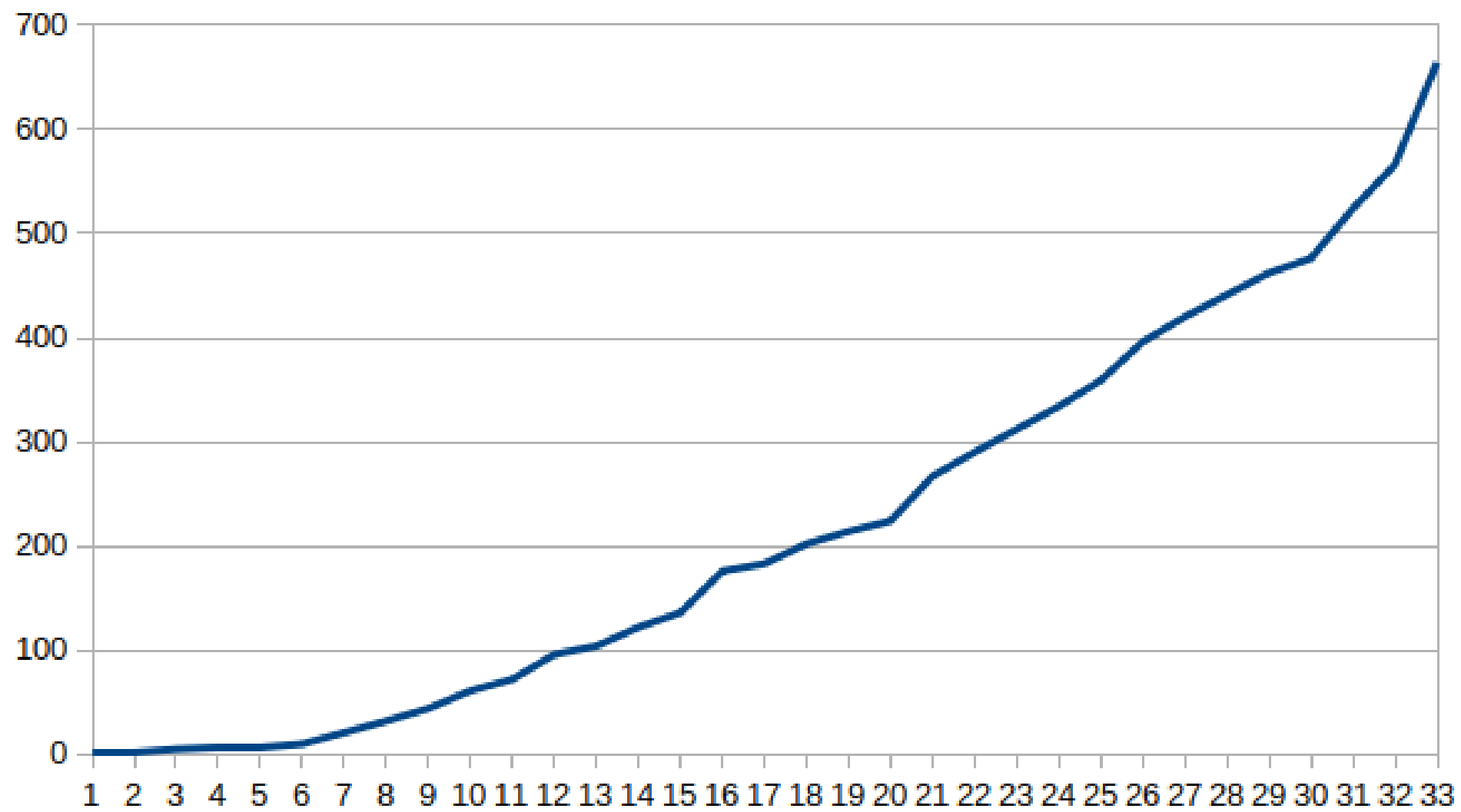
Log

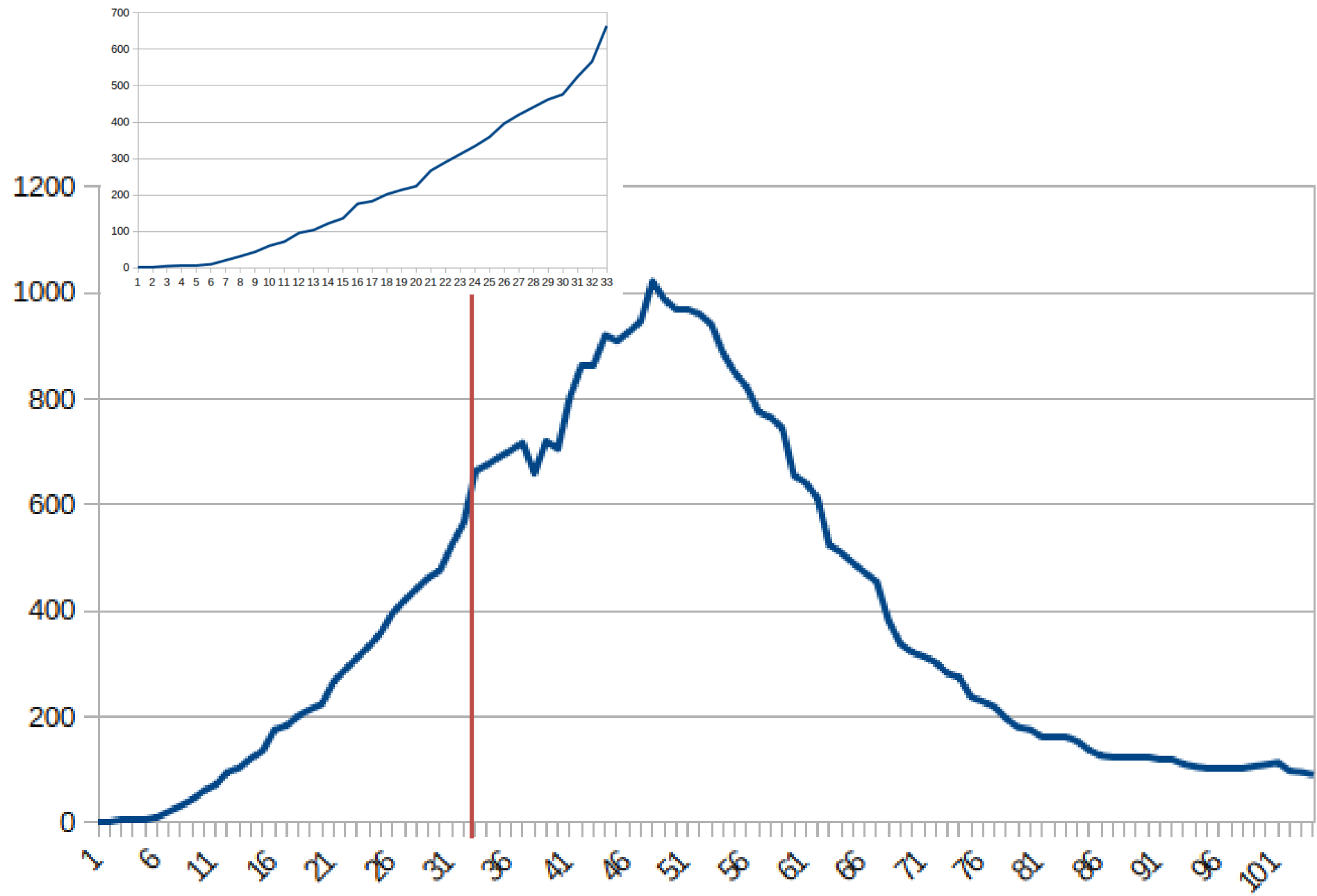


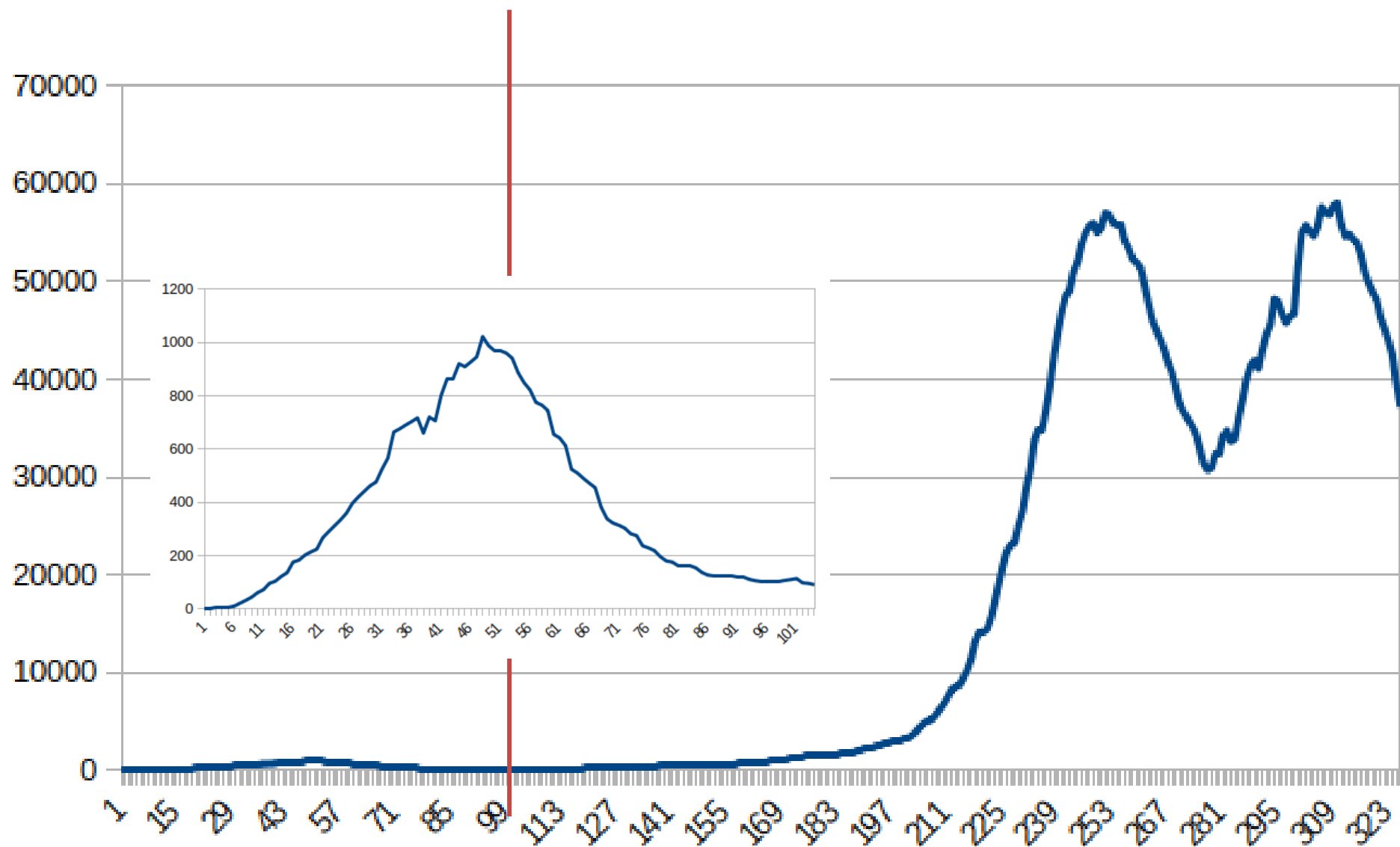
📅 11/03/2014 To 02/13/2023



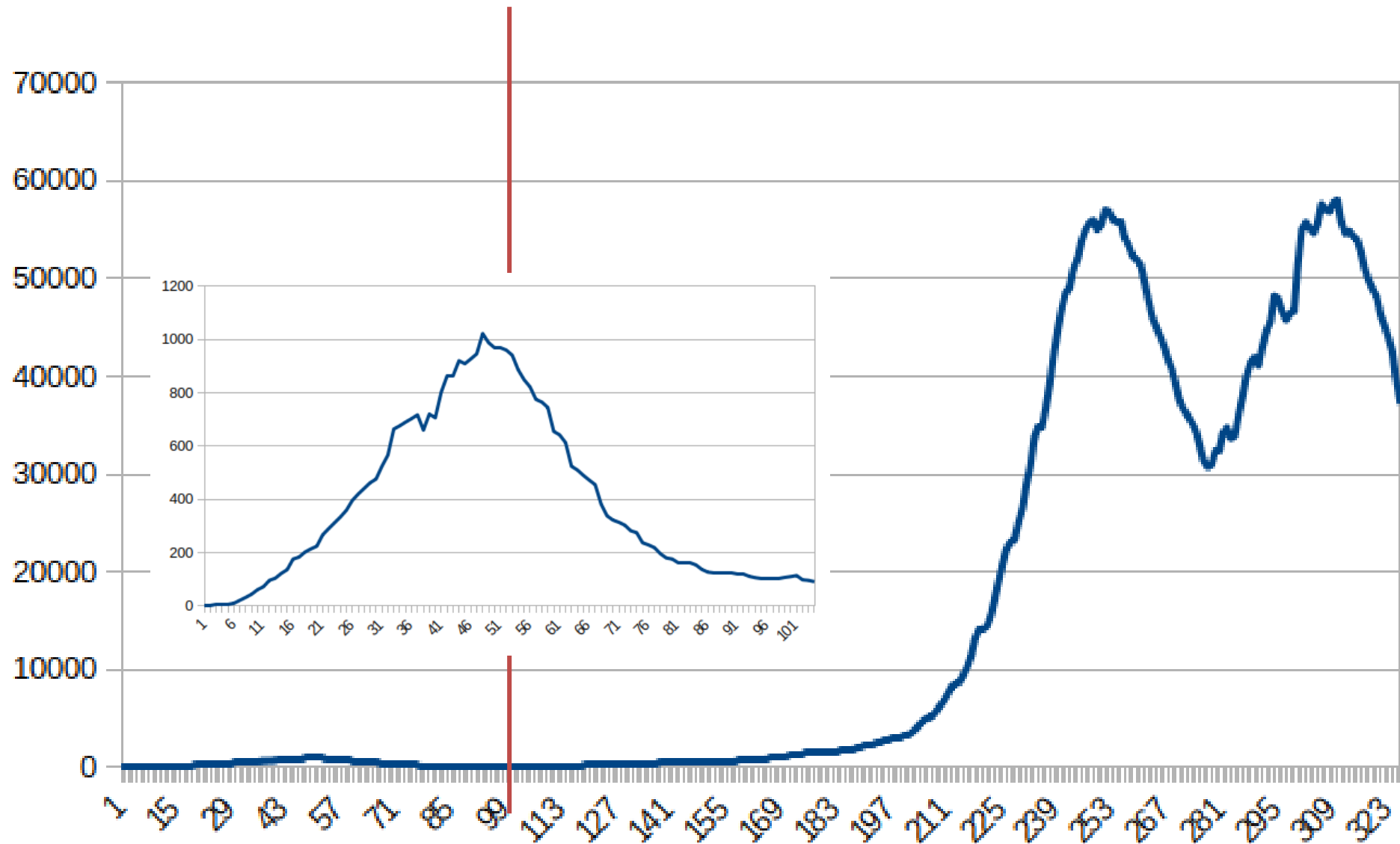
CoinDesk

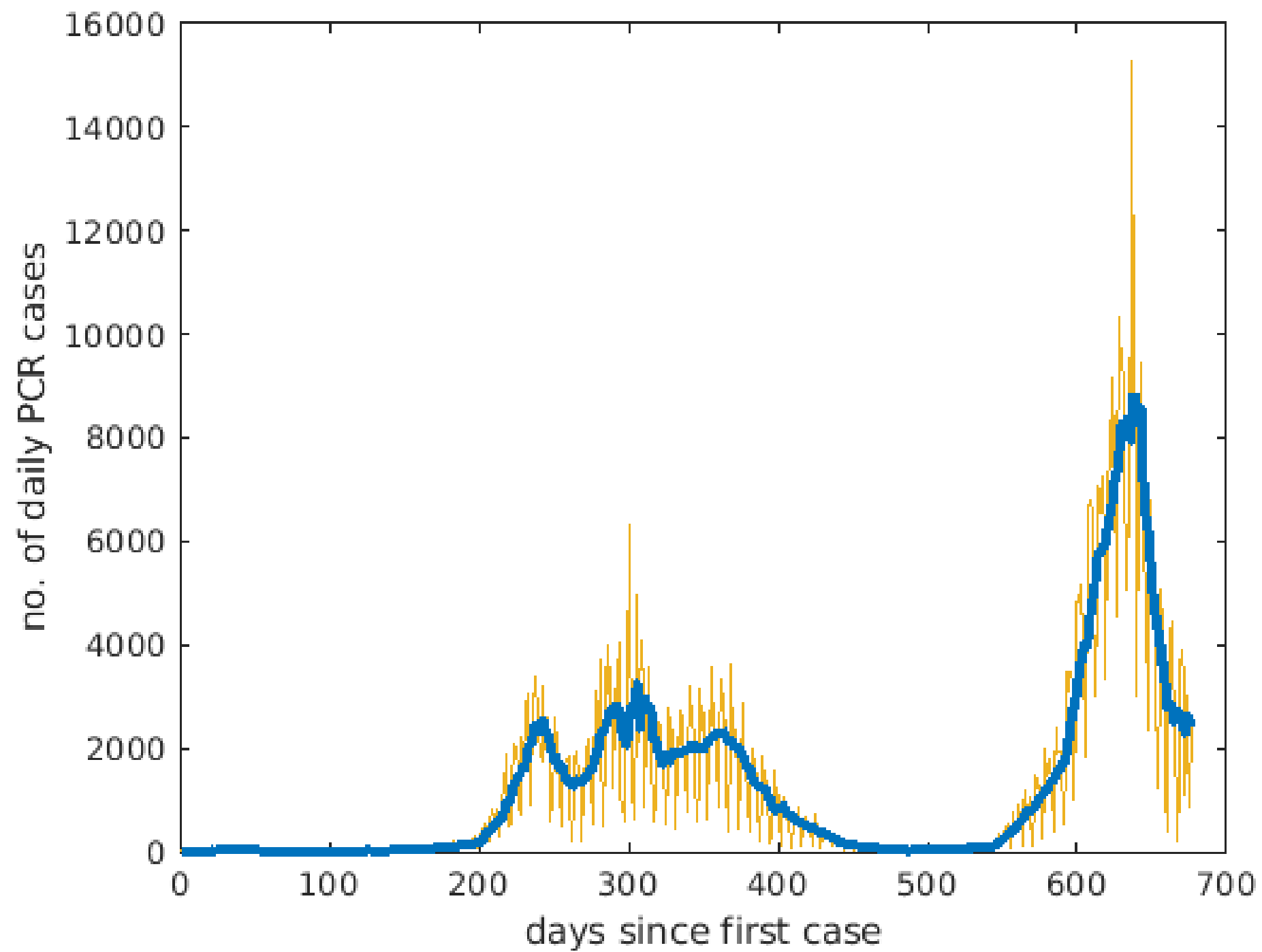




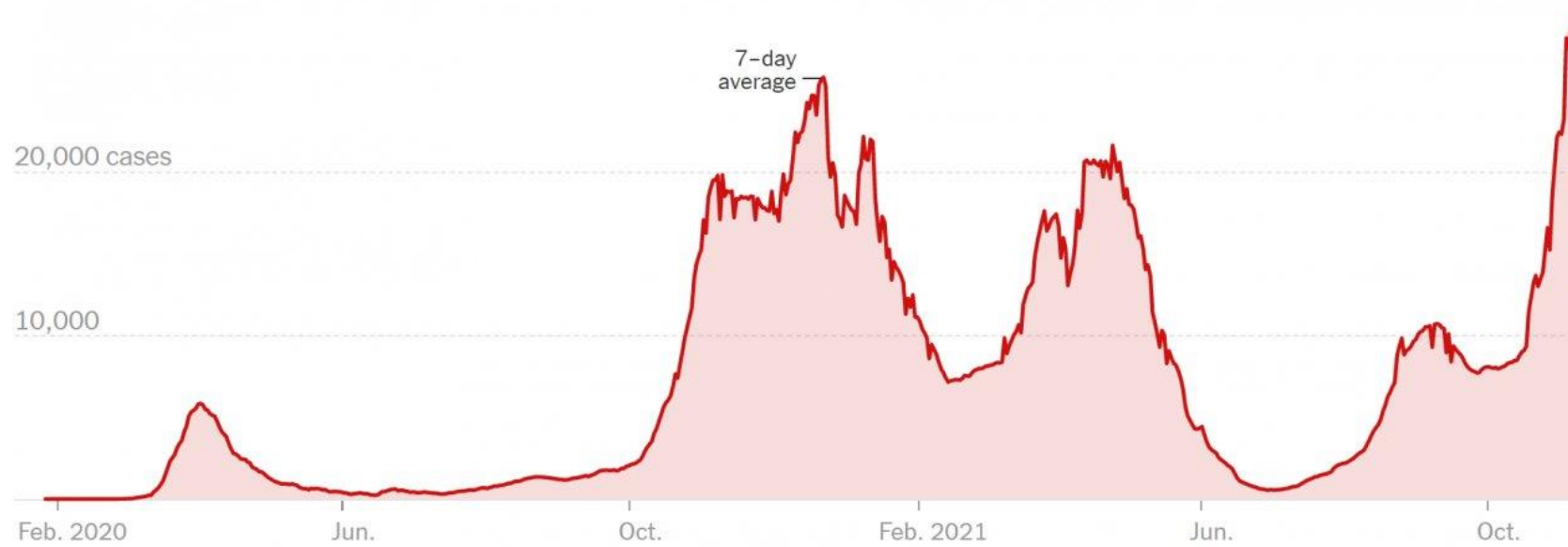


Active COVID-19 cases in Slovakia daily (March 2020 – January 2021)

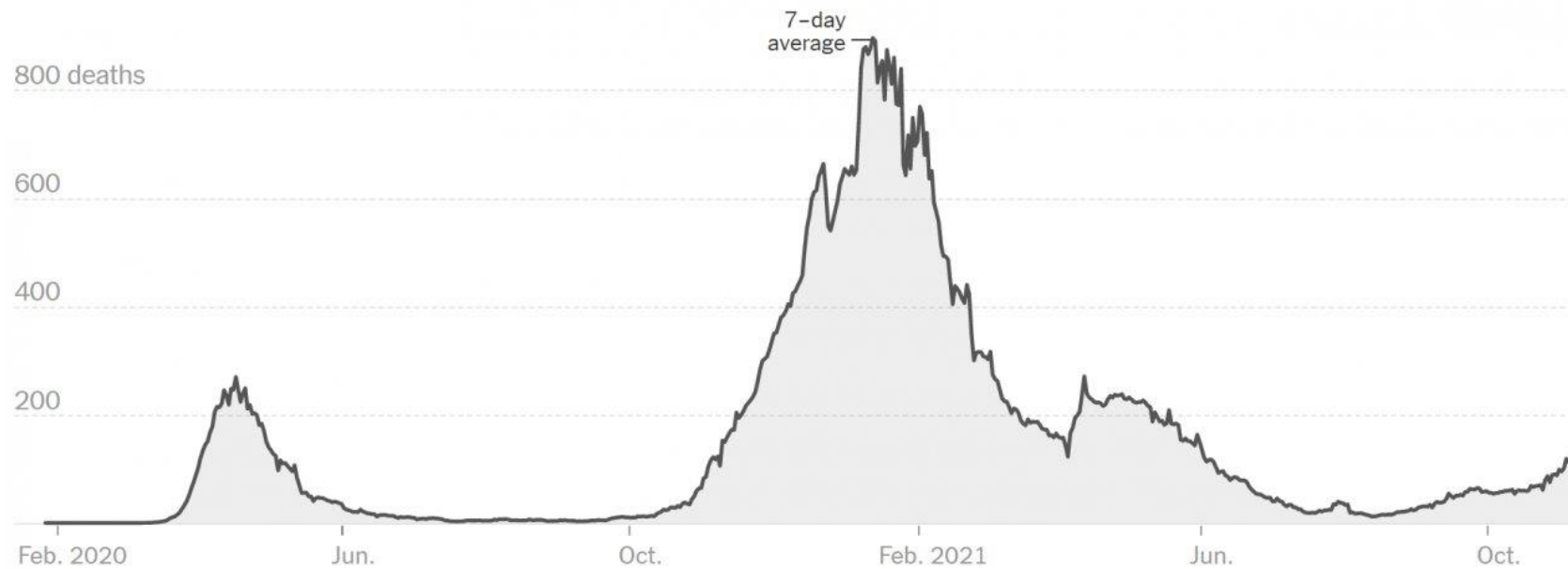


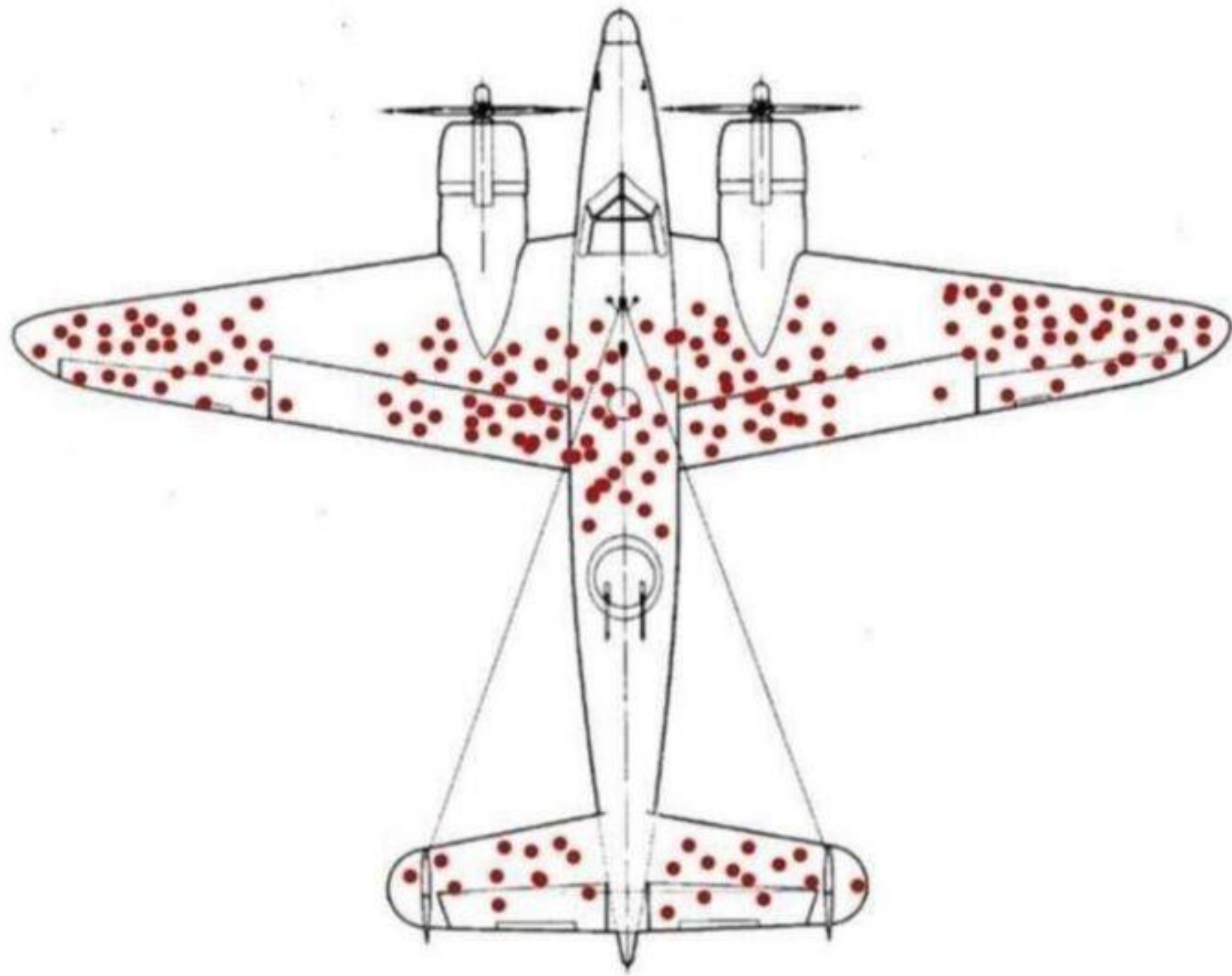


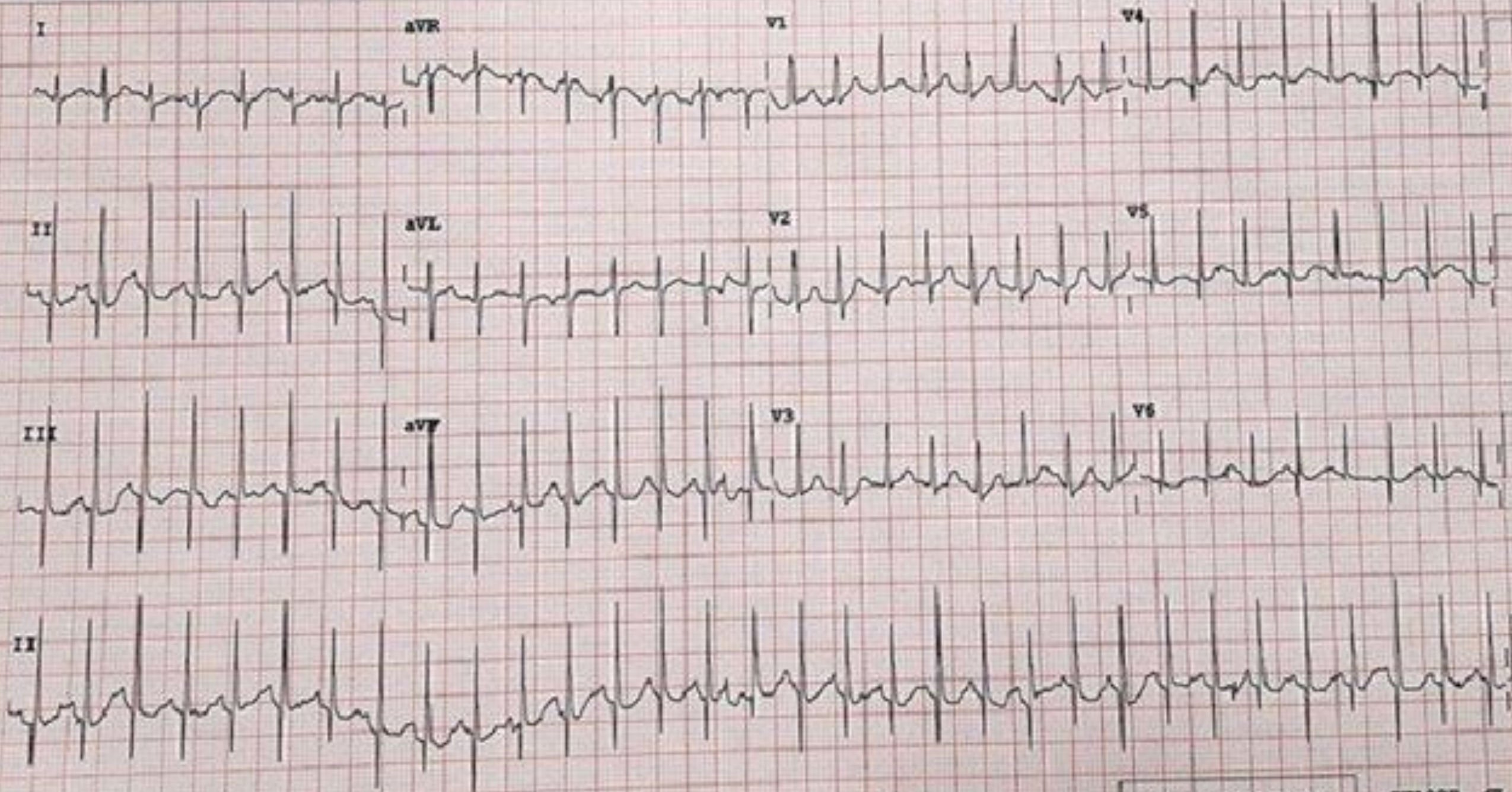
New reported cases by day



New reported deaths by day







Device:

Speed: 25 mm/sec

Limb: 10 mm/mV

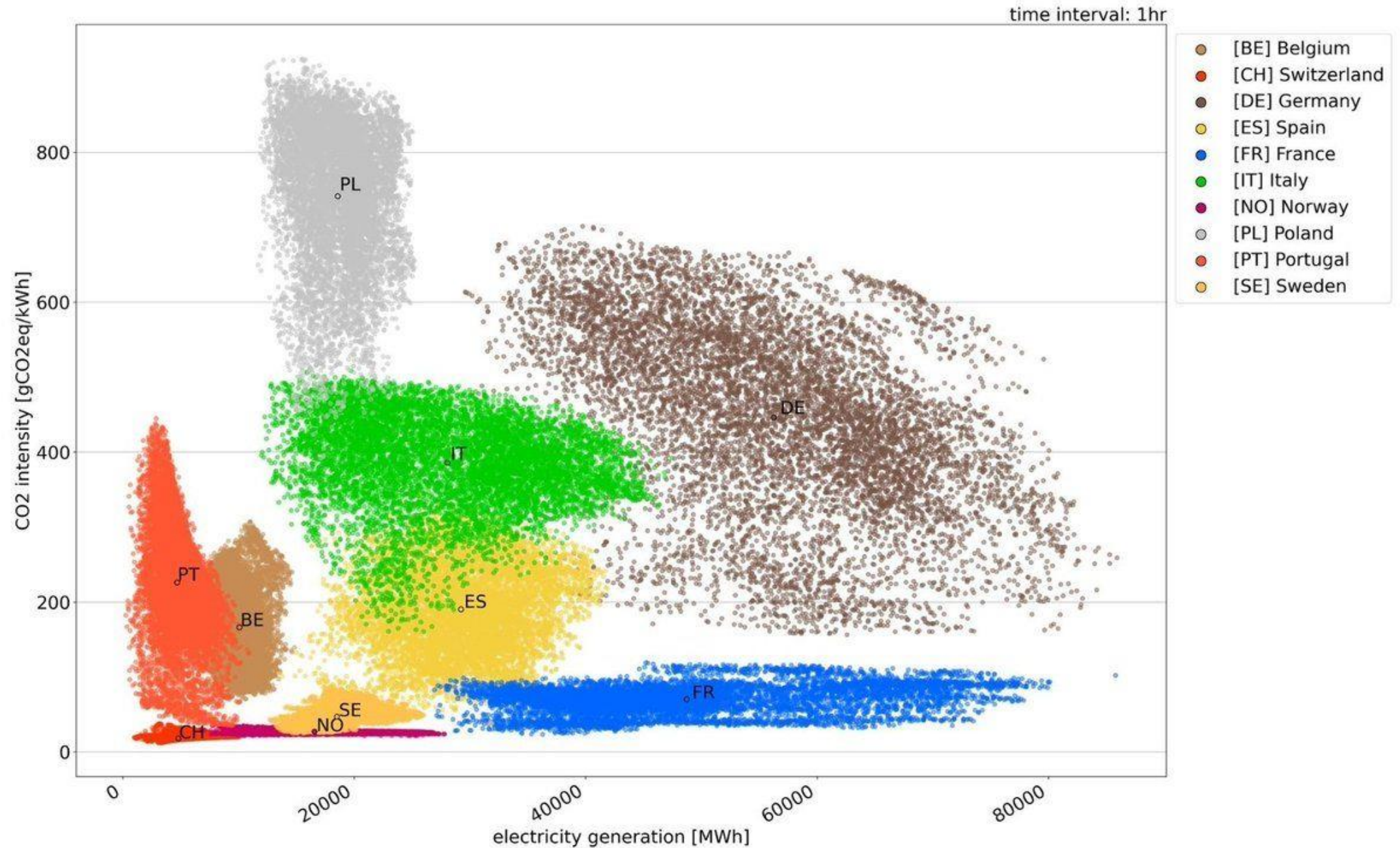
Chest: 10.0 mm/mV

F 60- 0.15-100 Hz

PH100B CL

Electricity generation compared to CO2 intensity per country
00:00 01/01/22 to 00:00 01/01/23

@BotElectricity v0.8
Source: ENTSO-E, IPCC 2014



Data visualization

- Time series
- Correlation plots
- Histograms
- Parity plots

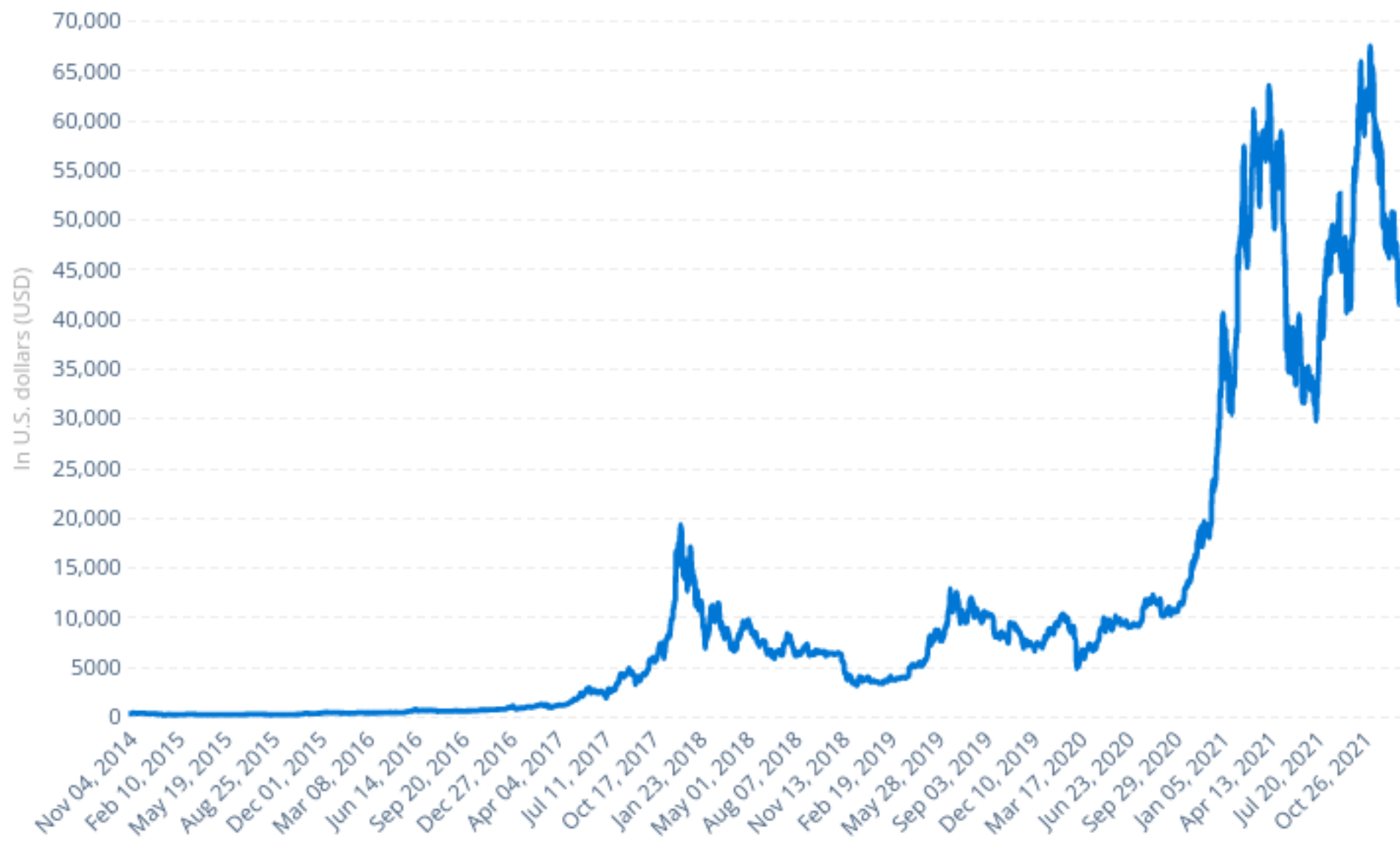
Data visualization

- Time series
- Correlation plots
- Histograms
- Parity plots

Bitcoin (BTC) price history

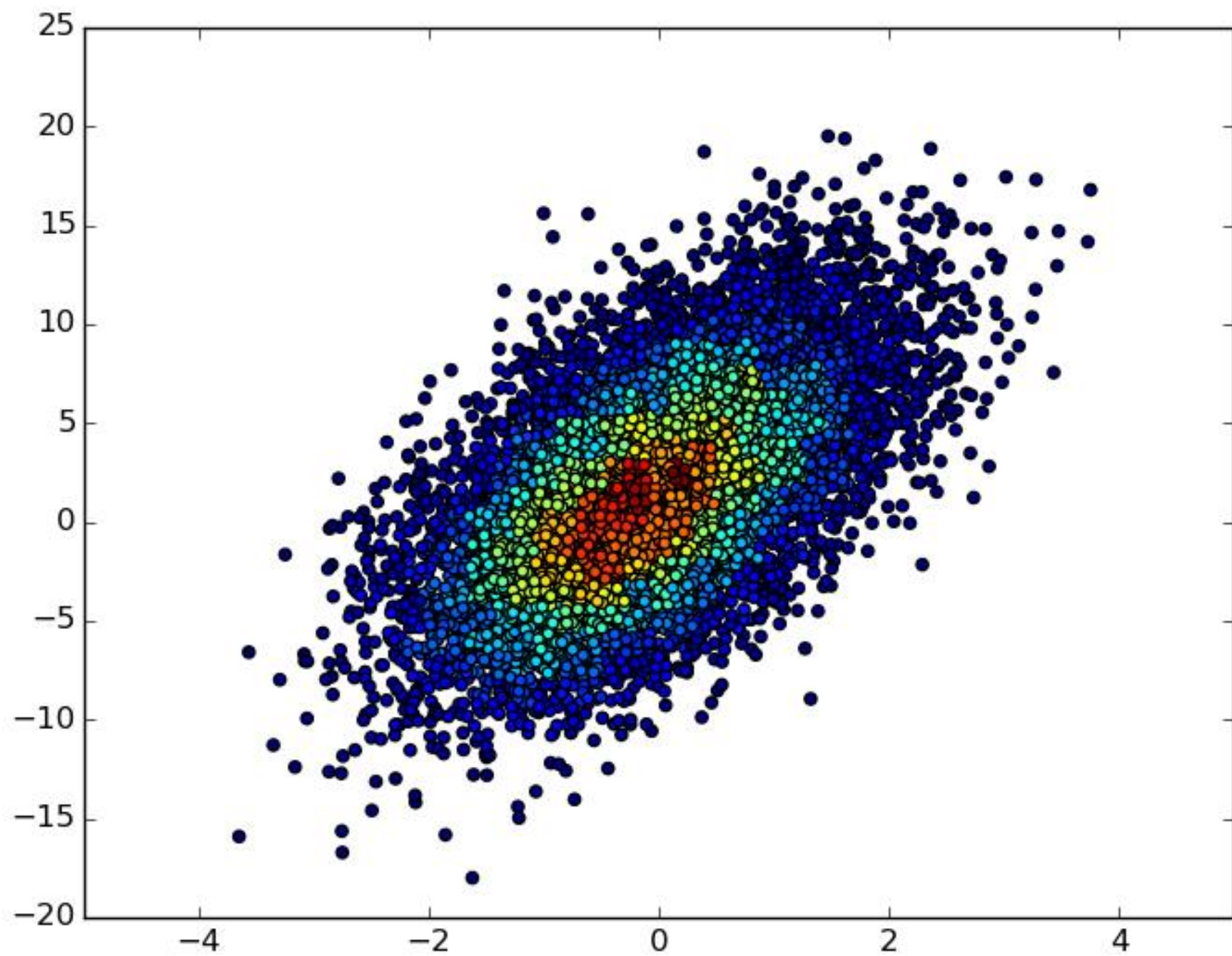
Linear

Log



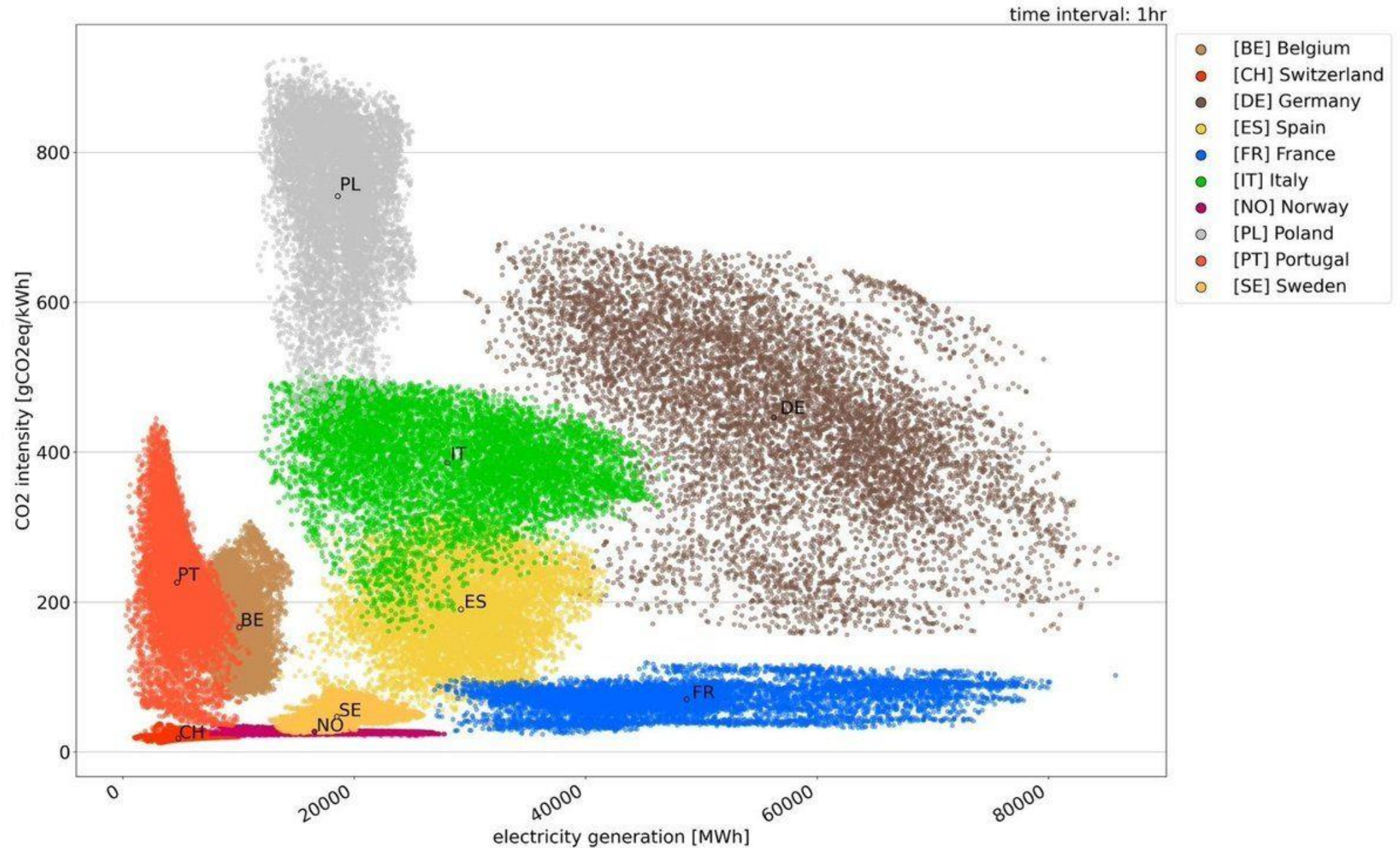
Data visualization

- Time series
- **Correlation plots**
- Histograms
- Parity plots



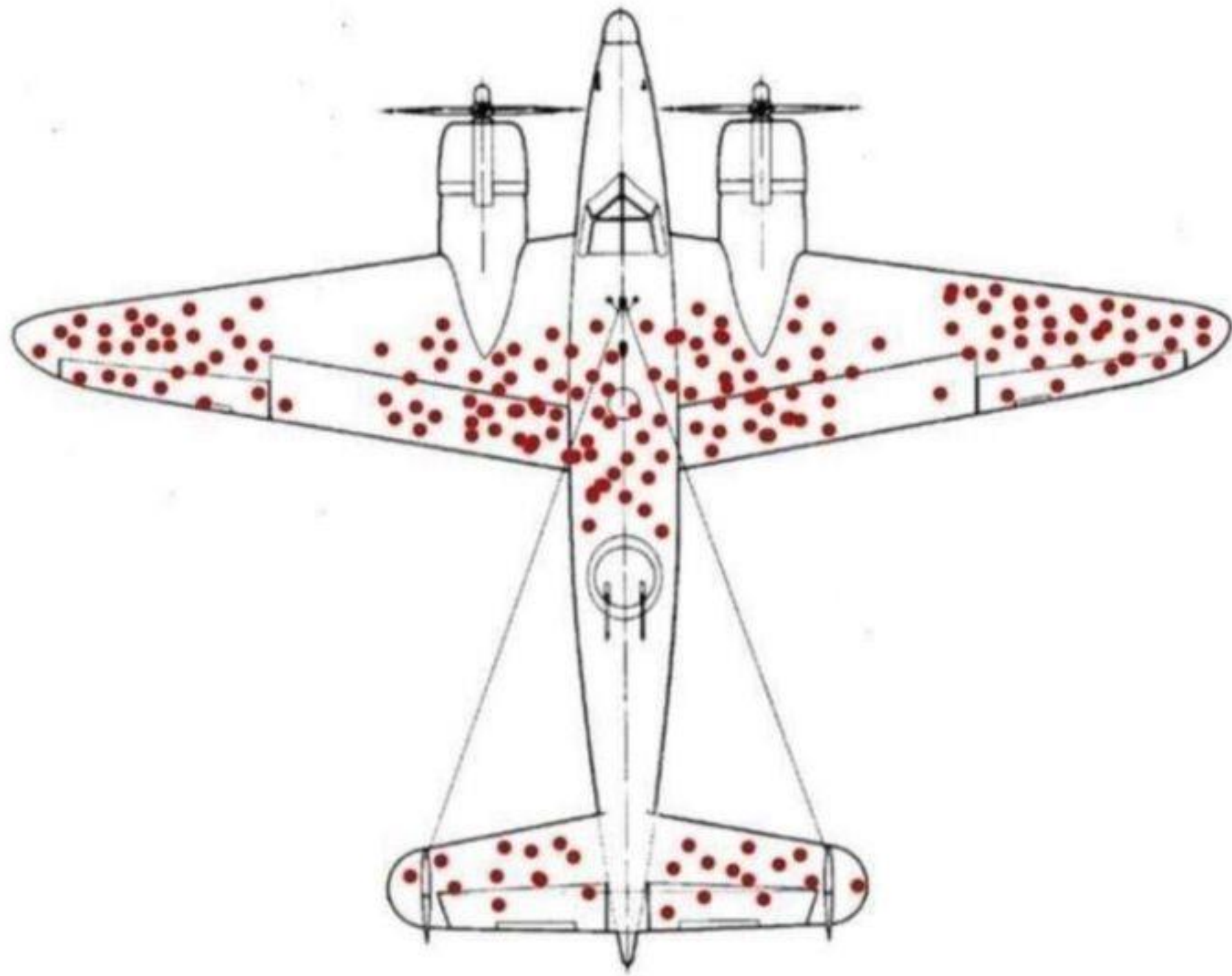
Electricity generation compared to CO2 intensity per country
00:00 01/01/22 to 00:00 01/01/23

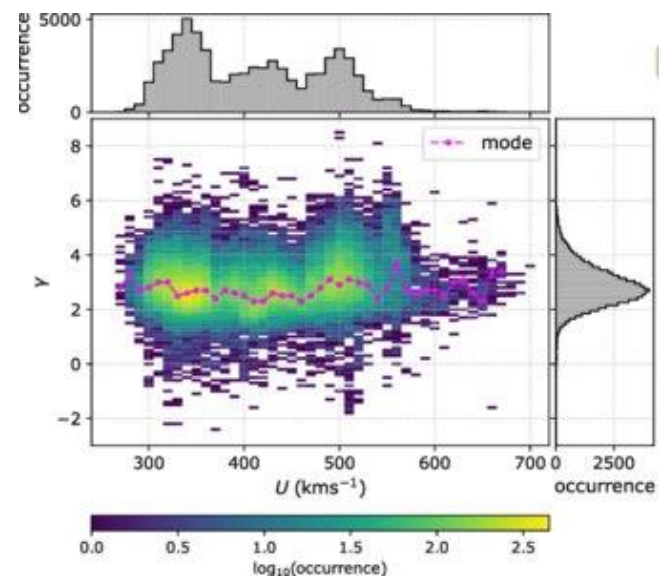
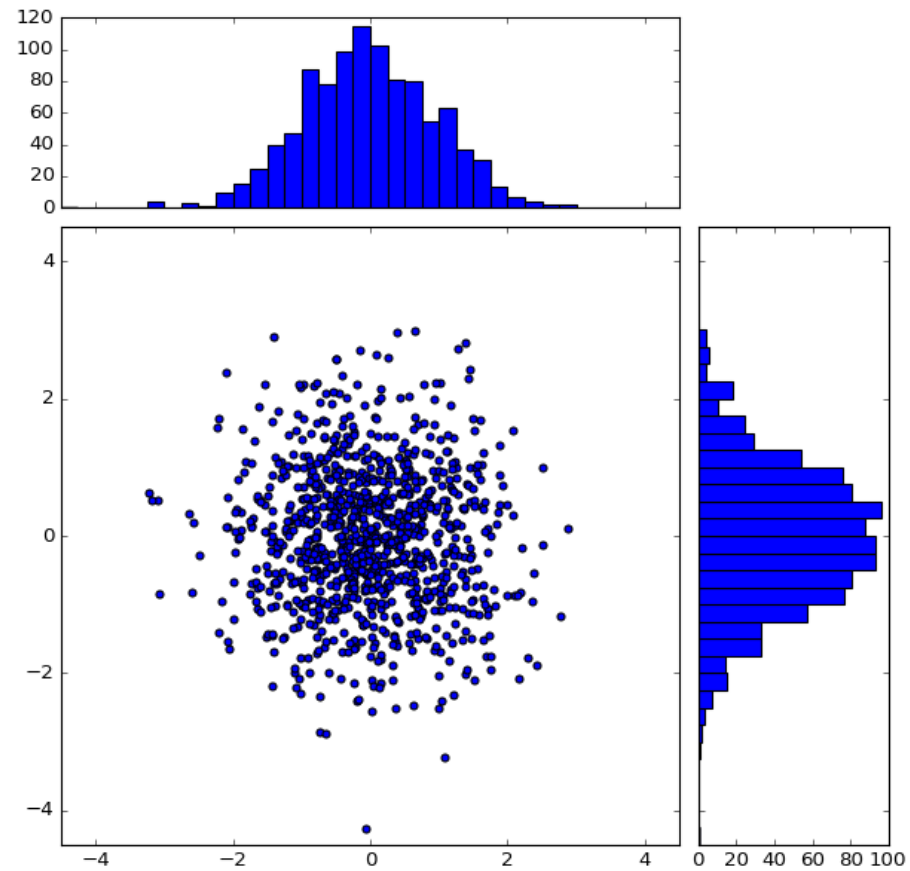
@BotElectricity v0.8
Source: ENTSO-E, IPCC 2014



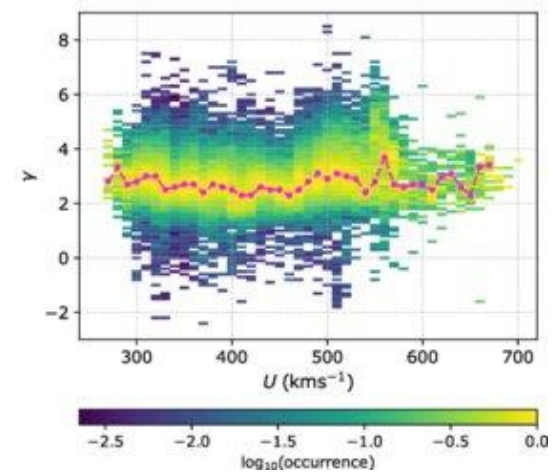
Data visualization

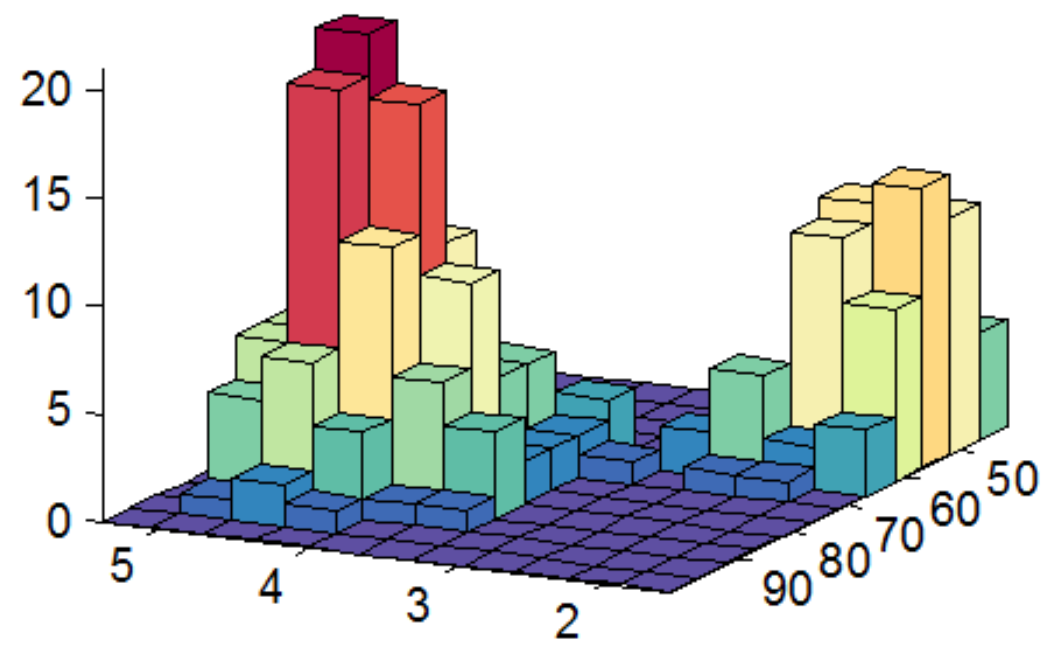
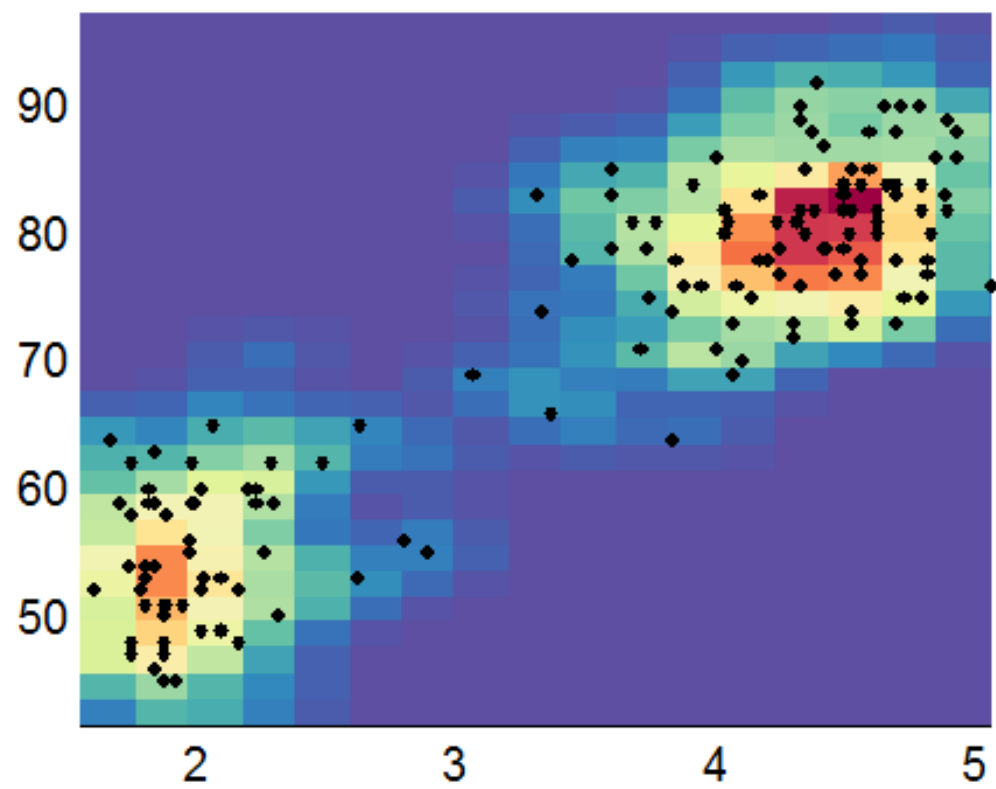
- Time series
- Correlation plots
- **Histograms**
- Parity plots





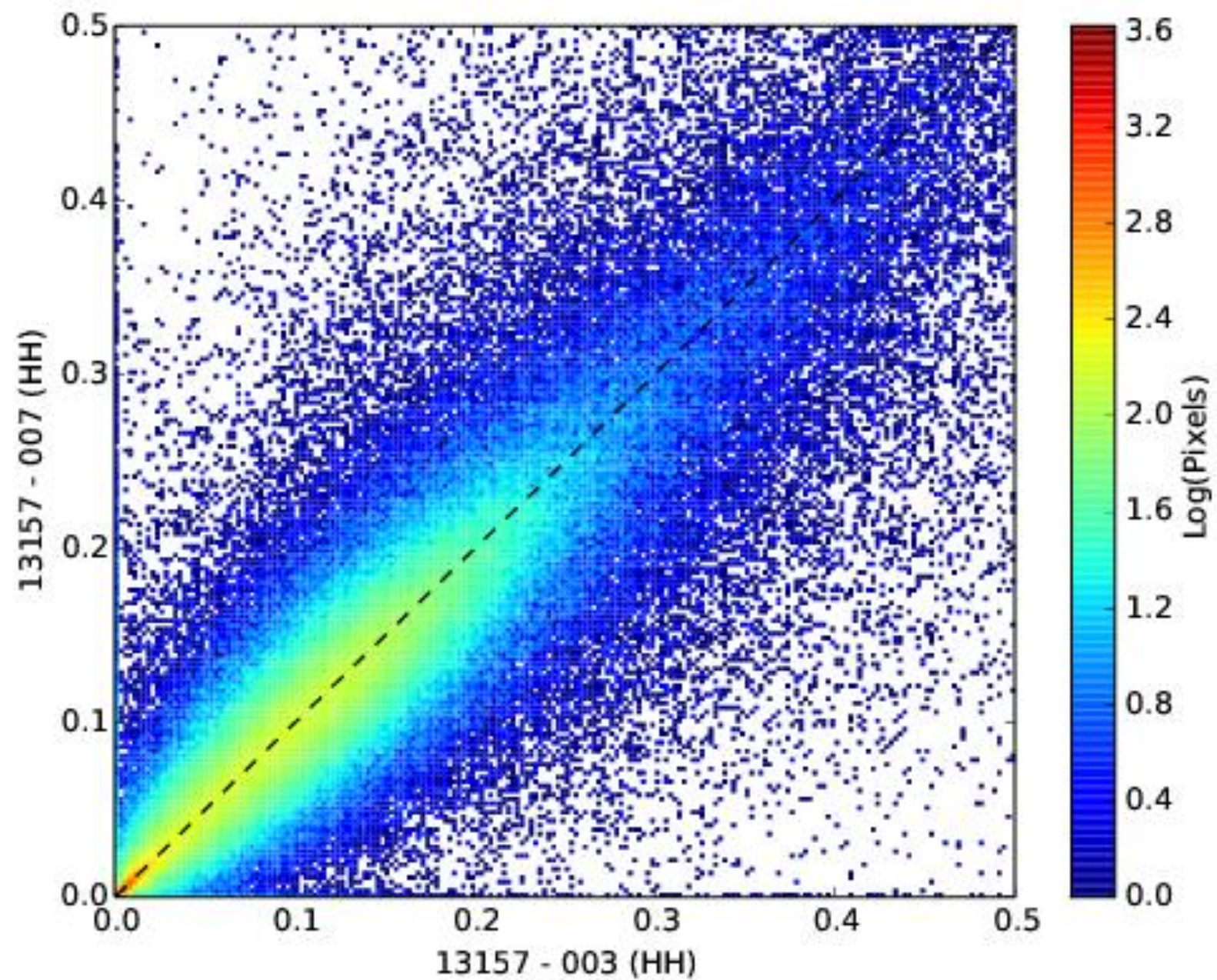
Interval 1





Data visualization

- Time series
- Correlation plots
- Histograms
- **Parity plots**

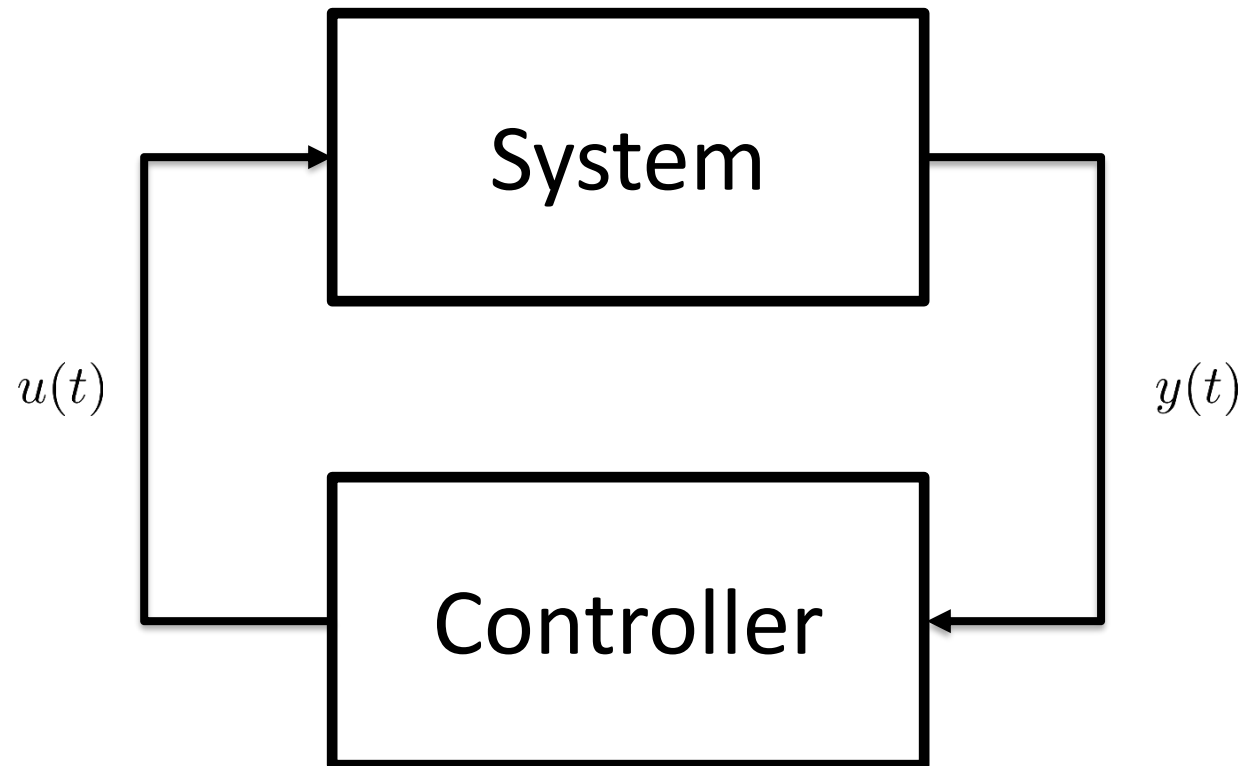


What is this course about?

- Identification -> System Identification
 - Finding mathematical models of dynamic systems using data
- Machine Learning
 - Finding mathematical models using data
- Data Science
 - Processing data and extracting information from it
- Statistics and Probability Theory
 - Modelling of uncertainty and randomness
- Filtering
 - Removing noise from data

What we need for “good” control?

- If we have a model of a controlled system, we can design and test/validate a controller.



What we need for “good” control?

- Model of a controlled system actually consists of:
 - Models of actuators
 - Models of physical sensors
 - Models of physical process behavior
 - Models of software sensors (inferential sensor, soft sensor)
 - Models of filters
- We need models for
 - Analysis, Simulation, Control, Optimization, Planning, Decision Making

Research (related to identification) at ÚIAM

- M. Kvasnica
 - Machine learning
- M. Klaučo
 - Machine learning
- J. Oravec
 - Dynamic model identification for control
- R. Paulen
 - Guaranteed and statistical estimation methods

Course literature

- M. Fikar: Identifikácia systémov. 1998
 - Available on Elearn/Moodle
- L. Ljung: System Identification: Theory for the User. 1999
 - From the author of System Identification toolbox in Matlab
- A.K. Tangirala: Principles of System Identification: Theory and Practice. 2015
 - Extensive, nice Matlab examples

We value your opinion

